

- 7 A piece of wire X and a negative temperature coefficient (NTC) thermistor are connected in series with a 9.0 V battery, as shown in Fig. 7.1.

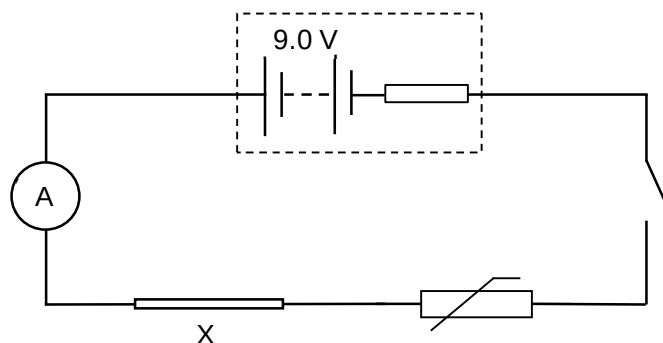


Fig. 7.1

- (a) When the switch is closed, the reading on the ammeter is 0.52 A.
- (i) Show that electrons are passing through any point in the circuit at a rate of $3.3 \times 10^{18} \text{ s}^{-1}$.

[1]

- (ii) Wire X has a diameter of 0.36 mm, and has a charge carrier number density of $5.9 \times 10^{28} \text{ m}^{-3}$.

Calculate the drift velocity of electrons in wire X.

drift velocity = m s^{-1} [2]

(iii) When the switch is closed, the current in the circuit flows almost immediately.

Explain this in the context of your answer to (ii).

.....

 [2]

(b) The I - V characteristics of wire X and the NTC thermistor are shown in Fig. 7.2.

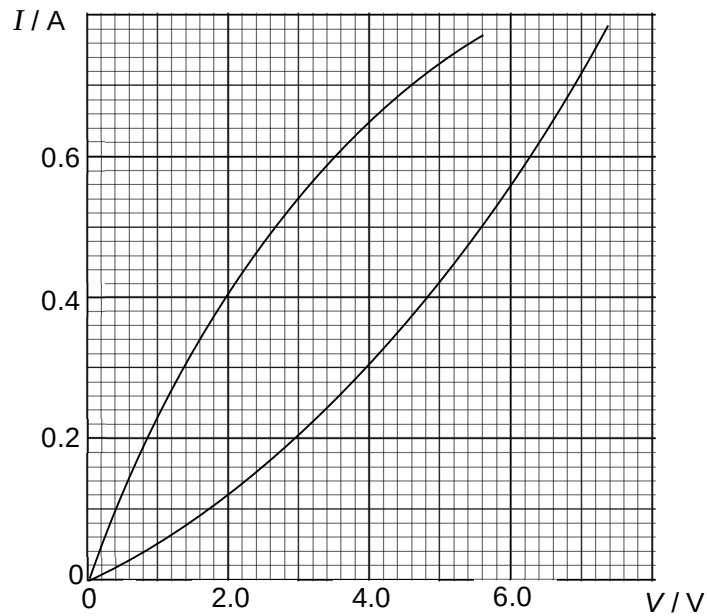


Fig. 7.2

Calculate the internal resistance of the battery.

internal resistance = Ω [2]

8 (a) In a nuclear reaction, a uranium-235 ($^{235}_{92}\text{U}$) nuclide is transformed into an unstable